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C03-EXP2. Dynamic Student Registration List Using DOM

Create a student registration interface to add, display, and remove student records dynamically. Use createElement(), appendChild(), remove(), parentElement, and children. Include Student Name, Register Number, Department, Add Student, Remove, and Clear All options.

PROGRAM:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Dynamic Student Registration</title>

    <style>

        * {

            box-sizing: border-box;

        }

        body {

            margin: 0;

            padding: 40px;

            font-family: Arial, sans-serif;

            background: linear-gradient(135deg, #dbeafe, #fce7f3);

        }

        .container {
```

```
max-width: 850px;
margin: auto;
background: white;
padding: 30px;
border-radius: 18px;
box-shadow: 0 8px 25px rgba(0, 0, 0, 0.15);
}
```

```
h1 {
  text-align: center;
  color: #1e3a8a;
  margin-bottom: 10px;
}
```

```
.subtitle {
  text-align: center;
  color: #64748b;
  margin-bottom: 25px;
}
```

```
.form-box {
  background: #fff6ff;
  padding: 20px;
  border-radius: 12px;
  margin-bottom: 25px;
}
```

```
.form-box input,
.form-box select {
  width: 100%;
```

```
padding: 12px;
margin: 8px 0;
border: 1px solid #cbd5e1;
border-radius: 8px;
font-size: 15px;
}
```

```
.form-box button {
width: 100%;
padding: 12px;
margin-top: 10px;
border: none;
border-radius: 8px;
background: #2563eb;
color: white;
font-size: 16px;
cursor: pointer;
}
```

```
.form-box button:hover {
background: #1d4ed8;
}
```

```
.list-header {
display: flex;
justify-content: space-between;
align-items: center;
}
```

```
.list-header h2 {
```

```
    color: #334155;
}
```

```
#studentCount {
    background: #dbeafe;
    color: #1d4ed8;
    padding: 6px 12px;
    border-radius: 20px;
}
```

```
#studentList {
    margin-top: 15px;
}
```

```
.student-card {
    display: flex;
    justify-content: space-between;
    align-items: center;
    background: #f8fafc;
    border-left: 5px solid #2563eb;
    padding: 15px;
    margin-bottom: 10px;
    border-radius: 10px;
}
```

```
.student-details {
    line-height: 1.6;
}
```

```
.student-details strong {
```

```
    color: #1e293b;
    font-size: 17px;
}
```

```
.student-details span {
    color: #64748b;
}
```

```
.remove-btn {
    background: #ef4444;
    color: white;
    border: none;
    padding: 9px 15px;
    border-radius: 7px;
    cursor: pointer;
}
```

```
.remove-btn:hover {
    background: #dc2626;
}
```

```
#clearAll {
    width: 100%;
    margin-top: 15px;
    padding: 12px;
    border: none;
    border-radius: 8px;
    background: #475569;
    color: white;
    font-size: 16px;
```

```
        cursor: pointer;
    }

    #clearAll:hover {
        background: #1e293b;
    }

    #emptyMessage {
        text-align: center;
        padding: 25px;
        color: #94a3b8;
    }
</style>
</head>

<body>

    <div class="container">

        <h1> 🎓 Student Registration</h1>

        <p class="subtitle">
            Add and manage student records dynamically using DOM
        </p>

        <!-- Registration Form -->
        <div class="form-box">

            <input
                type="text"
```

```
id="studentName"
placeholder="Enter Student Name"
>
```

```
<input
  type="text"
  id="registerNumber"
  placeholder="Enter Register Number"
>
```

```
<select id="department">
  <option value="">Select Department</option>
  <option value="Information Technology">Information Technology</option>
  <option value="Computer Science">Computer Science</option>
  <option value="Artificial Intelligence">Artificial Intelligence</option>
  <option value="Cyber Security">Cyber Security</option>
  <option value="Electronics and Communication">Electronics and
Communication</option>
</select>
```

```
<button onclick="addStudent()">
  + Add Student
</button>
```

```
</div>
```

```
<!-- Student List -->
```

```
<div class="list-header">
```

```
<h2>Registered Students</h2>
```

```
<span id="studentCount">
```

```
    Students: 0
```

```
</span>
```

```
</div>
```

```
<div id="studentList">
```

```
    <p id="emptyMessage">
```

```
        No students registered yet.
```

```
    </p>
```

```
</div>
```

```
<button id="clearAll" onclick="clearAllStudents()">
```

```
    Clear All Students
```

```
</button>
```

```
</div>
```

```
<script>
```

```
    // Add a new student
```

```
    function addStudent() {
```

```
        // Get input values
```

```
        let name = document.getElementById("studentName").value.trim();
```

```
        let registerNo = document.getElementById("registerNumber").value.trim();
```

```
        let department = document.getElementById("department").value;
```



```
// Validation
if (name === "" || registerNo === "" || department === "") {

    alert("Please fill all student details.");

    return;
}

// Get student list
let studentList = document.getElementById("studentList");

// Remove empty message
let emptyMessage = document.getElementById("emptyMessage");

if (emptyMessage) {
    emptyMessage.remove();
}

// Create student card
let studentCard = document.createElement("div");

studentCard.classList.add("student-card");

// Create student details section
let studentDetails = document.createElement("div");

studentDetails.classList.add("student-details");

// Create student name
let studentName = document.createElement("strong");
```

```
studentName.textContent = name;

// Create register number
let regText = document.createElement("div");

regText.textContent = "Register Number: " + registerNo;

// Create department
let deptText = document.createElement("div");

deptText.textContent = "Department: " + department;

// Add details to student details
studentDetails.appendChild(studentName);

studentDetails.appendChild(regText);

studentDetails.appendChild(deptText);

// Create remove button
let removeButton = document.createElement("button");

removeButton.textContent = "Remove";

removeButton.classList.add("remove-btn");

// Remove button functionality
removeButton.onclick = function () {
```

```
// parentElement finds the student card
let card = removeButton.parentElement;

// remove() removes the student card
card.remove();

updateCount();

checkEmptyList();
};

// Add button to student card
studentCard.appendChild(studentDetails);

studentCard.appendChild(removeButton);

// Add student card to student list
studentList.appendChild(studentCard);

// Clear input fields
document.getElementById("studentName").value = "";

document.getElementById("registerNumber").value = "";

document.getElementById("department").value = "";

// Update student count
updateCount();
}
```

```
// Update number of students
function updateCount() {

    let studentList = document.getElementById("studentList");

    // children gives all direct child elements
    let count = studentList.children.length;

    document.getElementById("studentCount").textContent =
        "Students: " + count;
}

// Check whether the list is empty
function checkEmptyList() {

    let studentList = document.getElementById("studentList");

    // children property checks existing student cards
    if (studentList.children.length === 0) {

        let message = document.createElement("p");

        message.id = "emptyMessage";

        message.textContent = "No students registered yet.";

        studentList.appendChild(message);
    }
}
```

```
// Clear all students
function clearAllStudents() {

    let studentList = document.getElementById("studentList");

    // Remove all student records
    while (studentList.children.length > 0) {

        studentList.children[0].remove();
    }

    // Add empty message
    let message = document.createElement("p");

    message.id = "emptyMessage";

    message.textContent = "No students registered yet.";

    studentList.appendChild(message);

    updateCount();
}
```

```
</script>
```

```
</body>
```

```
</html>
```

OUTPUT:



Student Registration

Add and manage student records dynamically using DOM

Select Department 

+ Add Student

Registered Students

Students: 2

savitha r

Register Number: 192421368

Department: Information Technology

Remove

raaja s

Register Number: 192421005

Department: Cyber Security

Remove

Clear All Students